

PROJECT PROPOSAL

Development of a Heterogeneous Aerial-Ground Swarm Robotics System for Search and Rescue

Senior Design Project

Project Name	Development of a Heterogeneous Aerial-Ground Swarm Robotics System for Search and Rescue
Faculty Advisor	Dr. Juan Calderon
Project Team Members	[Student Name 1] [Student Name 2] [Student Name 3] [Student Name 4] [Student Name 5] [Student Name 6]
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1. Project Name

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2. Project Team Members

Faculty Advisor: Dr. Juan Calderon

Student Team Members: [Student Name 1], [Student Name 2], [Student Name 3], [Student Name 4]

3. Problem Statement

3.1 What is the problem you are planning to solve?

Many robotic applications such as search and rescue, surveillance, exploration, transportation, security, and assistance to humans can benefit from the cooperation of multiple robots rather than the use of a single robotic platform. Aerial robots can rapidly observe large areas, obtain information from above, and reach locations that may be difficult for ground robots. Ground robots can operate close to objects and people, carry sensors or payloads, inspect areas at ground level, and typically remain operational for longer periods.

The objective of this project is to design, develop, and experimentally validate a complete heterogeneous swarm robotics system composed of an aerial robot swarm and a ground robot swarm. The project will begin with the development of the two swarms independently and will culminate in the integration of both swarms into a common cooperative architecture.

The ground swarm will consist of mobile robots, including spider-type legged robots and/or wheeled robots, controlled with ESP32 microcontrollers. Students will develop robot-level motion control, wireless communication between a central computer and multiple robots, a global localization system using an overhead camera and ArUco markers, closed-loop navigation, and multi-robot formation and swarm behaviors.

The aerial swarm will consist of Crazyflie drones. Students will install and configure the Lighthouse positioning system, establish communication with multiple Crazyflies, validate autonomous takeoff, landing, and waypoint motion, and develop aerial formation and swarm behaviors.

After both swarms operate independently, the project will address the more difficult integration problem: how to coordinate two swarms that have different dynamics, communication interfaces, localization technologies, control loops, dimensionality, and physical constraints. The final objective is a heterogeneous coordination layer that enables both swarms to cooperate during a simulated search-and-rescue mission.

3.2 Why is it a problem?

Swarm robotics studies how collective behavior can emerge from the interaction of multiple robots and is inspired in part by social insects and other biological collectives [1], [2]. In contrast with simple fleet control, swarm systems seek useful group behavior through coordination, local interaction, distributed decision making, or structured formation rules. Classical flocking research has also shown how relatively simple interaction rules can generate coherent collective motion [3].

Search-and-rescue operations are a strong application for multi-robot systems because different robotic platforms provide complementary capabilities. Surveys of multi-robot search and rescue identify coordination and interoperability among heterogeneous robots as important research challenges [4]. This is especially relevant when aerial and ground robots must cooperate while operating under different sensing,

mobility, communication, and control constraints. Previous research by the project advisor and collaborators has directly investigated swarm navigation and victim detection [10], bio-inspired navigation and victim localization [11], distributed particle-swarm-based search [12], visual victim detection with aerial sub-swarms and consensus [13], and leader-follower assistance behaviors for search and rescue [15].

In this project, the aerial and ground platforms are fundamentally different. Crazyflie drones move in three-dimensional space, require stabilized flight control, and will use Lighthouse for positioning. Ground robots primarily move in two dimensions, use ESP32-based embedded controllers, and will be localized by an external overhead camera using ArUco markers. Therefore, the same low-level control command cannot be directly applied to both platforms. The engineering problem is to create a common mission-level architecture that can coordinate these heterogeneous platforms while allowing each swarm to retain its own appropriate low-level controller.

A second important motivation is educational. Students usually encounter embedded systems, computer vision, networking, control, autonomous navigation, and robotics as separate topics. This project requires them to integrate these areas into a complete multi-robot system and then apply swarm-robotics theory to create collective behaviors.

3.3 Research Foundation and Continuity

The proposed Senior Design project is rooted in an established research line developed by the project advisor and collaborators. Earlier work introduced attraction-repulsion swarm navigation for search-and-rescue scenarios, including obstacle avoidance, victim localization, sub-swarm generation, and rendezvous-consensus formation around possible victims [10], [11]. Subsequent work extended these ideas through distributed particle swarm optimization for multi-robot search [12] and visual victim detection with CNN-based sensing and distributed estimation consensus in quadrotor swarms [13].

The same research program has also explored aerial leader-follower formation using fuzzy logic [14], PSO-based leader-follower behavior for robotic assistance in disaster zones [15], affordable and replicable physical ground swarm testbeds [16], cooperative deep reinforcement learning for aerial robot swarms operating under turbulent disturbances [17], and robust synchronization of interconnected heterogeneous quadrotors performing cooperative payload transportation [18]. The present project brings these complementary directions together in a hands-on educational platform in which students will build the ground and aerial swarms from the subsystem level and then integrate them into a heterogeneous aerial-ground robotic team.

3.4 Educational and Technical Learning Goals

- Swarm robotics: collective behavior, cooperation, emergence, scalability, local interaction, and distributed control concepts.
- Biologically inspired robotics: flocking, attraction-repulsion, cohesion, separation, alignment, leader following, and collective navigation.
- Formation control: line, wedge, circle, leader-follower, virtual-structure, consensus, potential-field, or another justified method.
- Computer vision: camera calibration, ArUco detection, robot identification, pose estimation, and coordinate transformation.
- Embedded systems: ESP32 programming, motor/actuator commands, communication protocols, and robot state management.
- Aerial robotics: Crazyflie configuration, Lighthouse positioning, autonomous flight, and multi-drone control.

- Systems integration: common robot representation, common coordinate frame, mission manager, data logging, and experimental validation.

3.5 Constraints

Physical Constraints

The aerial robots operate in three-dimensional space while the ground robots primarily operate in two dimensions. The two platforms have different velocities, accelerations, turning characteristics, stopping distances, and safety requirements. Laboratory size will limit the maximum swarm size and formation dimensions. The overhead camera must maintain visibility of the ArUco markers, and the Lighthouse system requires an appropriate indoor tracking volume and line of sight to the base stations [6].

Communication Constraints

Ground robots may use Wi-Fi, ESP-NOW, or another justified ESP32-compatible RF approach. Crazyflies use their own radio and software ecosystem. The integrated system must tolerate different update rates, command latency, packet loss, temporary connection failures, and simultaneous communication with multiple robots. The ESP32 platform supports IEEE 802.11 b/g/n Wi-Fi, which makes Wi-Fi a practical baseline option for the terrestrial swarm [8].

Localization Constraints

The ground and aerial swarms will initially use two independent localization systems. The ground swarm will use overhead computer vision with ArUco markers, while the aerial swarm will use Lighthouse. Students must establish a common mission coordinate frame so that the integrated system can reason about the relative positions of all robots. OpenCV provides ArUco detection and marker identification functions that are suitable for this global-vision subsystem [7].

Power Constraints

Both swarms operate from batteries. Flight endurance will limit the duration of Crazyflie experiments, while ground-robot batteries will constrain test duration, maximum speed, and repeated trials.

Safety Constraints

The system must prevent drone-drone, ground-ground, and aerial-ground collisions. Minimum separation distances, emergency-stop procedures, defined experimental boundaries, safe takeoff and landing, propeller safety, and battery-handling procedures must be included in the test plan.

Cost and Schedule Constraints

The project should reuse available Crazyflies, Lighthouse hardware, spider or wheeled robots, ESP32 boards, cameras, computers, and communication devices. The major schedule challenge is that students must build two independent robotic swarms before heterogeneous integration can begin. A staged development plan is therefore required.

4. Stakeholders

4.1 Primary Stakeholders

- Senior Design students, who will design, implement, integrate, test, document, and demonstrate the system.
- Faculty advisor/project sponsor, who will guide the technical direction and use the developed platform for education and future research.

- Future student teams and researchers, who can extend the platform with new swarm algorithms, sensors, missions, and robot types.

4.2 Secondary Stakeholders

- Search-and-rescue and emergency-response personnel, representing the principal application domain.
- Public-safety and security organizations interested in robotic surveillance and situational awareness.
- Robotics and autonomous-systems researchers.
- Organizations interested in logistics, exploration, inspection, transportation, or human assistance.
- The university and engineering program, which benefit from a reusable hands-on multi-robot educational platform.

5. Proposed Solution

The proposed solution is a staged heterogeneous swarm robotics architecture in which students develop the ground swarm and aerial swarm independently, validate each subsystem, and then integrate both using a common mission and coordination layer. The baseline search-and-rescue mission is an indoor simulation; surveillance, logistics, exploration, security, and human-assistance applications are treated as future extensions rather than required deliverables.

5.1 Stage 1 - Ground Swarm Development

Robot-Level Control

Students will program each ESP32-based mobile robot to execute reliable motion commands. Depending on the selected ground platform, commands may include forward motion, reverse motion, turning, stopping, heading control, and velocity control. Spider-type robots may also require gait-specific control.

Wireless Communication

Students will implement communication between a central computer and multiple ESP32 robots. The protocol should include robot identification, command type, desired motion or target position, robot state, acknowledgments, and emergency-stop functionality.

ArUco-Based Global Localization

An overhead camera will observe the ground-swarm workspace. Each robot will carry a unique ArUco marker. Students will calibrate the camera, detect the markers, identify each robot, estimate position and orientation, and convert image coordinates into physical world coordinates. ArUco markers are specifically designed for robust identification and pose-related computer-vision tasks [7].

Closed-Loop Navigation

The students will close the loop between desired target position, measured ArUco position, navigation error, and the ESP32 motion command. Individual waypoint navigation will be demonstrated before extending the controller to multiple robots.

Ground Swarm Behaviors

After reliable localization and navigation are achieved, students will implement and experimentally validate at least one justified swarm or formation-control approach. Candidate methods include leader-follower control, virtual structures, consensus, artificial potential fields, attraction-repulsion, and biologically inspired flocking. Initial geometric formations may include line, wedge, and circle formations. Prior work by the project advisor has shown how attraction-repulsion and distributed particle-swarm strategies can support

collective navigation, obstacle avoidance, and victim search [10]-[12], while the ARGroHBotS platform demonstrated the value of affordable, replicable physical testbeds for validating multi-agent and swarm algorithms [16].

5.2 Stage 2 - Aerial Swarm Development

Lighthouse Setup and Calibration

Students will install the Lighthouse base stations, configure the Crazyflies and Lighthouse decks, establish the Lighthouse geometry, define the coordinate system, and validate the usable flight volume. The Lighthouse system is an optical indoor positioning system designed for precise Crazyflie localization and requires proper geometry and line of sight [6].

Individual Crazyflie Control

Students will implement software to connect to a Crazyflie, read position and status, take off, land, send waypoint or velocity commands, and execute emergency-stop procedures.

Multi-Crazyflie Communication

The system will then be extended to multiple Crazyflies. Each drone will maintain a unique ID and state information such as position, target, connection status, and battery state. This provides the basis for aerial swarm coordination.

Aerial Swarm Behaviors

Students will implement aerial formation behaviors analogous to the ground-swarm experiments. The drones should be able to form a commanded geometry, maintain relative position, move while preserving formation, and, if time permits, transition between formations. Related work by the project advisor includes fuzzy-logic leader-follower formation for UAV swarms [14], PSO-based leader-follower behavior for search-and-rescue assistance [15], and cooperative deep reinforcement learning for scalable aerial-swarm trajectory tracking under turbulent disturbances [17]. These works provide candidate concepts for students to study while selecting an implementable controller for the Crazyflie platform.

5.3 Stage 3 - Heterogeneous Swarm Integration

After the two swarms function independently, students will develop a higher-level mission architecture that treats both swarms as components of one heterogeneous robotic system. The key design principle is separation between mission-level commands and platform-specific low-level control.

Common Robot Representation

The software should define a common robot abstraction containing fields such as robot ID, robot type, x-y-z position, orientation, velocity, state, target, and communication interface. High-level commands such as `move_to(target)` can then be translated by the aerial or ground interface into appropriate platform-specific commands.

Common Coordinate System

Because the ArUco and Lighthouse systems initially have different coordinate frames, students will derive coordinate transformations that express both aerial and ground robot positions in a common mission frame. This is essential for calculating relative geometry, heterogeneous formation error, and inter-platform safety distances.

Mission and Heterogeneous Coordination Layer

A mission manager will issue high-level objectives to the two swarm controllers. The aerial and ground swarms will retain their own low-level control loops while exchanging mission status and position information through the coordination layer. Prior research on interconnected heterogeneous quadrotors demonstrated distributed synchronization in the presence of different vehicle parameters, physical coupling, uncertainty, and disturbances [18]. The present project extends the heterogeneity challenge beyond aerial vehicles by coordinating platforms with different locomotion, localization, communication, and control architectures.

5.4 Bio-Inspired Swarm Component

The project is intended to teach swarm robotics rather than only multi-robot teleoperation. Students will therefore study and apply biologically inspired interaction principles such as separation, cohesion, alignment, attraction-repulsion, leader following, and local-neighbor interaction. Foundational swarm and flocking literature provides the theoretical basis for these concepts [1]-[3]. The project advisor's prior work provides additional application examples, including attraction-repulsion swarm navigation [10], [11], distributed PSO [12], fuzzy leader-follower formation [14], and PSO-based leader-follower coordination [15]. Students are not required to implement every method; they should compare candidate approaches and justify the selected algorithms.

5.5 Search-and-Rescue Demonstration Scenario

The demonstration scenario builds on prior research showing that robotic swarms can navigate through obstacles, maintain group behavior, create task-specific sub-swarms, and support victim localization and classification in simulated disaster environments [10]-[13]. The Senior Design implementation will focus on the physical swarm platforms and heterogeneous coordination rather than reproducing every perception capability from those studies.

- Initialize: both localization systems become operational and all robots report ready.
- Deploy: ground robots initialize and the Crazyflies take off.
- Form: each swarm establishes its commanded formation.
- Search: the aerial swarm moves ahead of or around the mission area to simulate rapid reconnaissance.
- Coordinate: the mission manager shares a target or rescue-zone objective with both swarms.
- Navigate: aerial and ground swarms move cooperatively toward the mission objective.
- Transform: one or both swarms change formation while maintaining safety spacing.
- Regroup: the heterogeneous team reaches the simulated rescue zone and reports mission completion.

5.6 In Scope

- ESP32 ground-robot programming and motion control.
- Wireless communication between a computer and multiple ground robots.
- Overhead camera setup, calibration, ArUco detection, and ground-robot localization.
- Closed-loop ground navigation and terrestrial swarm formation control.
- Crazyflie setup, Lighthouse installation and calibration, and autonomous flight control.
- Multi-Crazyflie communication and aerial swarm formation control.
- Study and implementation of selected biologically inspired swarm behaviors.
- A common coordinate frame and common robot software representation.
- Mission manager and heterogeneous coordination architecture.
- Experimental evaluation and a final indoor simulated search-and-rescue mission.

5.7 Out of Scope

- Outdoor drone deployment and outdoor FAA operational approval.
- Real disaster deployment or medical assessment of victims.
- Complex SLAM in unknown rubble environments.
- Deep-learning victim detection as a required deliverable. Prior work by the project advisor demonstrated CNN-based victim detection combined with aerial sub-swarm coordination and distributed estimation consensus [13], making vision-based victim recognition a well-grounded future extension after the core heterogeneous platform is validated.
- Heavy-payload transport or autonomous manipulation.
- Weather-resistant hardware.
- Large-scale swarms containing dozens or hundreds of robots.
- Complete decentralization of every sensing, planning, and control function.

5.8 Standards and Regulations

IEEE 802.11 is relevant if Wi-Fi is selected for ground-swarm communication. The ESP32 supports 2.4-GHz IEEE 802.11 b/g/n Wi-Fi [8]. Commercially certified radio modules should be used rather than designing custom RF transmitters. FCC Part 15 requirements are therefore relevant to the wireless hardware used in the United States.

The baseline aerial experiments are intended to be performed indoors. The FAA states that Part 107 does not apply to drone operations conducted indoors because indoor space is not part of the National Airspace System [9]. Any future outdoor deployment would require a separate review of applicable FAA requirements.

Laboratory safety procedures will address battery handling, flight boundaries, emergency stopping, personnel separation, propeller hazards, minimum robot spacing, and test authorization before multi-robot experiments.

5.9 System Functionalities by the End of the Project

- Identify and localize multiple ground robots using an overhead camera and unique ArUco markers.
- Send commands from a computer to multiple ESP32-controlled ground robots and receive state information.
- Navigate ground robots in closed loop to assigned target positions.
- Create and maintain at least one experimentally validated ground-swarm formation.
- Configure and localize multiple Crazyflies using Lighthouse.
- Command autonomous flight of multiple Crazyflies from a central computer.
- Create and maintain at least one experimentally validated aerial-swarm formation.
- Represent both swarms in one common coordinate frame.
- Issue mission-level commands through a common coordination architecture.
- Demonstrate cooperative aerial-ground behavior in an indoor search-and-rescue scenario.

6. Semester 1

Semester 1 will focus on foundational learning, hardware/software setup, localization, communication, individual closed-loop control, and initial homogeneous swarm behavior.

1. Study swarm robotics, multi-agent coordination, bio-inspired control, and formation-control literature.
2. Program and validate ESP32 ground-robot motion control.

3. Develop computer-to-ground-robot wireless communication.
4. Install and calibrate the overhead camera and implement ArUco localization.
5. Implement closed-loop waypoint navigation for at least one ground robot and then multiple robots.
6. Demonstrate an initial ground-swarm formation if schedule permits.
7. Configure Crazyflies and install/calibrate the Lighthouse positioning system.
8. Implement individual Crazyflie autonomous takeoff, waypoint motion, and landing.
9. Expand the software to communicate with multiple Crazyflies.
10. Demonstrate initial coordinated aerial motion and, if schedule permits, an initial aerial formation.
11. Implement data logging for robot position, commands, timing, and experimental metrics.

Minimum Semester 1 demonstration: at least two ground robots demonstrate closed-loop navigation using ArUco localization, while at least two Crazyflie drones demonstrate autonomous localization and controlled flight using Lighthouse. Initial formation behavior should be demonstrated on at least one of the two platforms.

7. Semester 2

Semester 2 will focus on completing both homogeneous swarms, creating the common software and coordinate architecture, integrating the two swarms, and experimentally validating the heterogeneous system.

1. Complete and validate ground-swarm formation generation, maintenance, motion, and transitions.
2. Complete and validate aerial-swarm formation generation, maintenance, motion, and transitions.
3. Develop a common mission coordinate frame for ArUco and Lighthouse measurements.
4. Develop a common robot abstraction and platform-independent mission command interface.
5. Implement the heterogeneous mission manager and coordination layer.
6. Demonstrate shared trajectory following and relative aerial-ground geometry.
7. Evaluate robustness to selected disturbances such as displacement, communication delay, temporary localization degradation, or robot dropout.
8. Execute and document the final simulated search-and-rescue mission with both swarms operating cooperatively.

8. Engineering Performance Metrics

The evaluation plan is informed by the advisor's previous studies of swarm navigation, formation, and robustness. Relevant precedents include navigation and victim-localization performance [10]-[13], explicit formation-error evaluation in leader-follower systems [15], scalable aerial-swarm tracking under disturbances [17], and robust synchronization under uncertainty [18]. The project will adapt these ideas into measurable hardware-based metrics for both homogeneous swarms and the final heterogeneous mission.

Metric	Description
Localization Error	Difference between measured robot position and a reference or known position for both ArUco and Lighthouse subsystems.
Formation Error	Difference between desired robot positions in the formation and measured positions; RMS error can be used across all agents.
Convergence Time	Time required to establish a commanded formation after initialization or a formation change.

Safety Spacing	Minimum measured separation and number of safety-spacing violations.
Trajectory Tracking Error	Difference between desired mission trajectory and actual swarm trajectory.
Communication Latency	Measured delay between command transmission and robot response or acknowledgment.
Command Success Rate	Percentage of transmitted commands that are correctly received and executed.
Formation Transition Time	Time required to transition from one formation geometry to another.
Mission Success Rate	Successful end-to-end mission trials divided by total trials.
Robustness	Change in performance after selected disturbances, communication delays, localization errors, or robot dropout.

9. Proposed Project Budget

The principal robotic platforms and laboratory equipment are expected to be available, including Crazyflie drones, Lighthouse positioning hardware, ESP32-based ground robots, an overhead camera, computing equipment, chargers, and wireless communication hardware. Therefore, no major capital purchase is required for the baseline project. A small contingency budget is recommended for replacement batteries, spare propellers, robot components, ArUco mounting materials, cables, chargers, electronic components, and hardware damaged during iterative testing.

Recommended contingency budget: \$500-\$1,000, subject to an equipment inventory before the project begins. If the course proposal is submitted under the rubric option that permits no required budget, the team may instead state that the core project requires no additional budget and use existing laboratory resources.

10. References

- [1] E. Şahin, "Swarm Robotics: From Sources of Inspiration to Domains of Application," in *Swarm Robotics, Lecture Notes in Computer Science*, vol. 3342, Springer, 2005, pp. 10-20. doi: 10.1007/978-3-540-30552-1_2.
- [2] M. Brambilla, E. Ferrante, M. Birattari, and M. Dorigo, "Swarm Robotics: A Review from the Swarm Engineering Perspective," *Swarm Intelligence*, vol. 7, no. 1, pp. 1-41, 2013. doi: 10.1007/s11721-012-0075-2.
- [3] R. Olfati-Saber, "Flocking for Multi-Agent Dynamic Systems: Algorithms and Theory," *IEEE Transactions on Automatic Control*, vol. 51, no. 3, pp. 401-420, 2006. doi: 10.1109/TAC.2005.864190.
- [4] J. Peña Queralta et al., "Collaborative Multi-Robot Search and Rescue: Planning, Coordination, Perception, and Active Vision," *IEEE Access*, vol. 8, pp. 191617-191643, 2020. doi: 10.1109/ACCESS.2020.3030190.
- [5] C. W. Reynolds, "Flocks, Herds and Schools: A Distributed Behavioral Model," *Computer Graphics*, vol. 21, no. 4, pp. 25-34, 1987. doi: 10.1145/37401.37406.
- [6] Bitcraze AB, "Lighthouse Positioning System" and "Getting Started with the Lighthouse System," Crazyflie Documentation, accessed Sep. 2026.
- [7] OpenCV, "Detection of ArUco Markers," OpenCV Documentation, version 4.x, accessed Sep. 2026.

- [8] Espressif Systems, “ESP32 Series Datasheet,” version 5.3, 2026. Wi-Fi support: IEEE 802.11 b/g/n.
- [9] Federal Aviation Administration, “Do the FAA rules and regulations apply to a commercial UAS or drone operations conducted indoors ONLY?,” FAA UAS FAQ, accessed Sep. 2026.

References [10]-[18] are prior publications co-authored by the project advisor and directly related to the technical foundations or future extensions of this project.

- [10] G. A. Cardona and J. M. Calderon, “Robot Swarm Navigation and Victim Detection Using Rendezvous Consensus in Search and Rescue Operations,” *Applied Sciences*, vol. 9, no. 8, art. 1702, 2019. doi: 10.3390/app9081702.
- [11] J. Leon, G. A. Cardona, A. Botello, and J. M. Calderon, “Robot Swarms Theory Applicable to Seek and Rescue Operation,” in *Intelligent Systems Design and Applications, Advances in Intelligent Systems and Computing*, vol. 557, Springer, 2017, pp. 1061-1070. doi: 10.1007/978-3-319-53480-0_104.
- [12] D. Paez, J. P. Romero, B. Noriega, G. A. Cardona, and J. M. Calderon, “Distributed Particle Swarm Optimization for Multi-Robot System in Search and Rescue Operations,” *IFAC-PapersOnLine*, vol. 54, no. 4, pp. 1-6, 2021. doi: 10.1016/j.ifacol.2021.10.001.
- [13] G. A. Cardona, J. Ramirez-Rugeles, E. Mojica-Nava, and J. M. Calderon, “Visual Victim Detection and Quadrotor-Swarm Coordination Control in Search and Rescue Environment,” *International Journal of Electrical and Computer Engineering*, vol. 11, no. 3, pp. 2079-2089, 2021. doi: 10.11591/ijece.v11i3.pp2079-2089.
- [14] W. O. Quesada, J. I. Rodriguez, J. C. Murillo, G. A. Cardona, D. Yanguas-Rojas, L. G. Jaimes, and J. M. Calderon, “Leader-Follower Formation for UAV Robot Swarm Based on Fuzzy Logic Theory,” in *Artificial Intelligence and Soft Computing, Lecture Notes in Artificial Intelligence*, vol. 10842, Springer, 2018, pp. 740-751. doi: 10.1007/978-3-319-91262-2_65.
- [15] N. Gomez, N. Pena, S. Rincon, S. Amaya, and J. Calderon, “Leader-Follower Behavior in Multi-Agent Systems for Search and Rescue Based on PSO Approach,” in *SoutheastCon 2022*, pp. 413-420, 2022. doi: 10.1109/SoutheastCon48659.2022.9764133.
- [16] N. I. Ospina, E. Mojica-Nava, L. G. Jaimes, and J. M. Calderon, “ARGroHBotS: An Affordable and Replicable Ground Homogeneous Robot Swarm Testbed,” *IFAC-PapersOnLine*, vol. 54, no. 13, pp. 256-261, 2021. doi: 10.1016/j.ifacol.2021.10.455.
- [17] D. Patino, S. Mayya, J. Calderon, K. Daniilidis, and D. Saldana, “Learning to Navigate in Turbulent Flows with Aerial Robot Swarms: A Cooperative Deep Reinforcement Learning Approach,” *IEEE Robotics and Automation Letters*, vol. 8, no. 7, pp. 4219-4226, 2023. doi: 10.1109/LRA.2023.3280806.
- [18] G. A. Cardona, M. F. Arevalo-Castiblanco, D. Tellez-Castro, J. M. Calderon, and E. Mojica-Nava, “Robust Adaptive Synchronization of Interconnected Heterogeneous Quadrotors Transporting a Cable-Suspended Load,” in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 31-37, 2021. doi: 10.1109/ICRA48506.2021.9561513.